

HIDDEN SLIDE H1 – WHY WE'RE SO CHEAP

Visual: The hardware BOM / cost breakdown (see backend/findings/power-budget.md and backend/findings/costs.md), plus the "install in 5 min" line.

Title on slide: "Cheap at every layer: hardware, connectivity, power, data, and coverage."

| BULLET POINTS ON SLIDE

- **Hardware:** ~€24 BOM per node at volume. Standard, reusable off-the-shelf sensors.
- **Install in 5 minutes.** Stake it next to the tree, register it in the app, and it joins the mesh and knows where it is. No grid hookup, no trenching, no specialist.
- **Connectivity:** ~€1 per node. ESP-NOW mesh is free between nodes; only the central gateway nodes carry WiFi, amortised to roughly €1 per node across the cluster. No SIM, no LoRaWAN fee, no WiFi per tree.
- **Power: €0 running cost.** Solar + LiPo, no grid, no battery swaps, no maintenance truck rolls.
- **Tiny energy budget.** The node deep-sleeps and wakes ~every 15 min with the sensors power-gated off, drawing ~7 µA: on the order of **0.2 mAh/day (~60 mAh/year)** in the production design. A 5×5 cm solar cell is overkill, so it runs for years with no battery swap. *(The hackathon dev board draws more through its USB chip; the custom PCB hits these numbers.)*
- **Data: free.** Open-Meteo / DWD weather costs nothing.
- **Coverage: nearly free.** The hybrid model covers all **126k trees** from a few hundred sensors, so we don't pay for a sensor per tree.
- **Rewards: existing city capacity,** not cash out the door (seed packets, museum entry, transit pass).

| WHAT TO SAY

- "The cost isn't just a cheap sensor. Every single layer is cheap: the build, the radio, the power, the weather data, and the coverage."
- "Installing one takes about five minutes. Stake it, register it in the app, and it joins the mesh and knows exactly where it is."
- "Connectivity is roughly a euro per node. Only the gateways talk WiFi, and they cover a whole cluster."
- "Power is basically free. It sleeps at a few microamps, wakes every 15 minutes to read and send, and a tiny 5-by-5 cm solar panel covers it many times over. The production design runs for years on one battery."
- "A whole one-district pilot is about €1,200 of hardware. One young tree that dies costs five to fifteen thousand to replace."

| WHY A CITIZEN USES IT

- Cheap enough that the city can actually deploy widely, so there are real thirsty trees near you to rescue.
- Low running cost keeps the program alive, so the rewards keep coming.

| WHY THE CITY USES IT

- Fits a tight, legally-mandated budget: low total cost of ownership, no per-tree capex.
- Pays for itself against a single avoided tree death; scales without a sensor per tree.

Visual: Screenshot from the route planner / city interface (e.g. the noise view).

Title on slide: "A multi-truck route planner, on a live per-tree model of the city."

| BULLET POINTS ON SLIDE

- **Multi-truck route planning.** The hybrid model ranks every tree by water stress; the planner splits the thirsty trees across multiple trucks into efficient routes, instead of fixed loops that miss the trees actually drying out.
- **Risk triage.** Young, high-stress trees rise to the top of the list; healthy mature ones drop off it.
- **Turn-by-turn for citizens.** The app routes a volunteer to the nearest tree they can actually help.
- **A live per-tree, per-street model underneath.** Soil, microclimate, heat, and activity, ground-truthed by sensors. Not a generic weather grid.
- **Proof that an intervention worked.** Before/after on real continuous data: did pedestrianising this street, or Tempo-30, actually cool it or cut the noise?
- **Evidence the city can defend.** Numbers for the Klimaanpassungsgesetz report and for budget decisions, instead of "we think it helped."
- **Closed loop:** plan, act, verify, re-plan.

| WHAT TO SAY

- "Instead of fixed routes that miss the trees actually drying out, the planner takes the ranked thirsty trees and splits them efficiently across your trucks."
- "Underneath it is a living, per-tree, per-street model of the city, anchored to real sensors."
- "Same data proves interventions: did pedestrianising this street, or Tempo-30, actually do what it promised, on continuous data instead of a one-off survey?"

| WHY A CITIZEN USES IT

- One decision: the app sends you to the nearest tree you can save, with a route, not a list.

| WHY THE CITY USES IT

- Multi-truck planning over the trees that actually need water: less labour, fewer misses than a fixed loop.
- A standing model instead of one-off studies, with evidence to defend interventions and compliance reporting.

Visual: Pull screenshots from the report (Section 5.2 — depth/coverage and the dense/sparse split).

Title on slide: "The probe reads the top 5 cm. Physics models the rest of the tree, and most of the city."

| BULLET POINTS ON SLIDE

- **A sensor only reads the top ~5 cm** of soil, but the roots drink 20–30 cm down, which can be dry while the surface looks fine.
- **Physics-based vapourisation model.** For every tree we don't actively measure, we estimate its water loss (evapotranspiration) from the **weather forecast** plus **data from nearby measured trees**.
- **So a few sensors cover many trees.** Measured trees ground-truth the model; the model fills in the rest.
- **Young trees: dense coverage.** High risk, they die without help, and citizen watering is verifiable on them.
- **Old trees: sparse coverage.** Deep roots, they live through most things, so we interpolate and barely sensor them.
- **Honest caveat:** constants are physical estimates today, not yet fitted to real dry-down data; it calibrates as sensors deploy.

| WHAT TO SAY

- "The probe only sees the top five centimetres. The tree drinks much deeper, so we never trust the surface alone."
- "For trees without a sensor, a physics model estimates their water loss from the weather forecast and from the trees nearby that do have sensors."
- "That's why young trees get dense sensors — high risk, and citizens can actually help — and old trees get almost none, because they survive most things and the model covers them."

| WHY A CITIZEN USES IT

- The thirst rating is real physics plus nearby ground-truth, so the tree you're sent to genuinely needs water.

| WHY THE CITY USES IT

- Whole-cadastre coverage from sparse hardware, concentrated where the risk and the verifiable saves actually are.